

Polished Dataset Stage 1 Smoke Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Summary

The `polished_dataset_stage1_smoke_2026_06_21` campaign completed the approved Stage 1 smoke matrix on 2026-06-22 with 8 completed runs and 0 failed runs in the accepted execution attempt.

The scalar winner is `te_periodic_gru_sequence_remote_global`, trained on `polished_dataset` with the `polished_point_v1` schema:

- test MAE: 0.001279 deg ;
- test RMSE: 0.001638 deg ;
- validation MAE: 0.001274 deg ;
- trainable parameters: 157,569 .

This is a normal campaign closeout. It accepts the Stage 1 training smoke matrix, updates the scalar registries, and records the polished-dataset artifact state. It does not promote a new official compensation candidate through the `TE Curve Verification Pipeline`; that curve-first verification remains a separate operator-approved workflow.

Campaign Scope

Stage 1 was intentionally small. Its purpose was to prove that representative repository-owned model families can train end-to-end on the new `polished_dataset` point schema before preparing larger retraining waves.

The campaign exercised these model branches

- feedforward baseline;
- tabular tree benchmark;
- harmonic regression baseline;
- periodic GRU sequence model;
- residual harmonic GRU sequence model;
- curve-aware full-curve composite model;
- harmonic-prior residual model;
- latent-state hysteresis GRU offset-residual model.

All accepted runs used the polished input columns:

Field	Role
<code>theta</code>	motor position measured in degrees
<code>theta_dot</code>	motor velocity derived from position
<code>tau_load</code>	applied load in Nm
<code>T</code>	oil temperature
<code>theta_TE</code>	measured transmission error target

The campaign-level leaderboard and family registry metadata were checked after completion. The tree run was confirmed to have trained on `polished_dataset` from its real `metrics_summary.yaml` ; a legacy registry fallback initially reported the tree entry as `simplified_dataset` . The aggregation code and generated registry artifacts were repaired so the tree entry now records `polished_dataset` , `polished_point_v1` , and input dimension `4` .

Execution Attempts

Attempt	Start	Outcome	Closure Handling
first launch	2026-06-22 11:55:01	failed during feedforward best-checkpoint reload	archived as diagnostic evidence; fixed <code>input_size: auto</code> reload handling
second launch	2026-06-22 12:46:43	interrupted during feedforward after dataloader bottleneck warning	archived before relaunch; enabled automatic dataloader worker sizing
accepted launch	2026-06-22 13:09:57	completed all <code>8</code> queued runs	accepted for Stage 1 closeout

The accepted campaign output directory is `output/training_campaigns/2026-06-22-13-09-57_polished_dataset_stage1_smoke_2026_06_21`

Scalar Leaderboard

Identifiers in this table are shortened for PDF readability. The exact `run_name` , `run_instance_id` , checkpoint, metrics snapshot, and training report path are recorded in `output/training_campaigns/2026-06-22-13-09-57_polished_dataset_stage1_smoke_2026_06_21/campaign_leaderboard.yaml` .

Rank	Run	Family	Test MAE	Test RMSE	Val MAE	Params
1	<code>periodic_gru_global</code>	<code>periodic_gru_sequence</code>	0.001279	0.001638	0.001274	157,569
2	<code>hist_gbr_global</code>	<code>tree</code>	0.001753	0.002892	0.001591	4
3	<code>track2g_full_curve_global</code>	<code>track2g_full_curve_composite</code>	0.002008	0.002581	0.001872	85,440
4	<code>residual_harmonic_gru_sparse_global</code>	<code>residual_harmonic_gru_sparse</code>	0.002112	0.002699	0.001978	150,676
5	<code>harmonic_prior_smooth_l1_global</code>	<code>harmonic_prior_residual</code>	0.002168	0.002763	0.001889	7,168
6	<code>track2h_gru_offset_global</code>	<code>track2h_gru_offset_residual</code>	0.002339	0.002986	0.002232	124,899
7	<code>feedforward_trial</code>	<code>feedforward</code>	0.002877	0.003835	0.002725	26,113
8	<code>harmonic_order12_global</code>	<code>harmonic_regression</code>	0.003839	0.004555	0.003904	125

Run Duration Summary

Run	Duration
te_feedforward_trial	00:03:27
te_hist_gbr_tabular_global	00:01:53
te_harmonic_order12_linear_conditioned_recovery_global	00:11:27
te_periodic_gru_sequence_remote_global	00:40:03
te_residual_harmonic_gru_sequence_remote_global_sparse_rcim	00:24:21
te_track2g_curve_aware_full_curve_composite_global	00:44:57
te_wave3_harmonic_prior_residual_smooth_l1_structured_global	00:21:31
te_track2h_l_gru_offset_residual_global	00:23:03

Registry And Artifact Effects

The accepted run updated family-level registries for:

- feedforward ;
- tree ;
- harmonic_regression ;
- periodic_gru_sequence ;
- residual_harmonic_gru_sequence_sparse_rcim ;
- track2g_curve_aware_harmonic_residual_offset_full_curve_composite_global ;
- wave3_harmonic_prior_residual_smooth_l1_structured_global ;
- track2h_latent_state_hysteresis_gru_offset_residual_global .

The program scalar registry now points to te_periodic_gru_sequence_remote_global as the current scalar best by test MAE. This supersedes the previous scalar registry winner but does not replace the direction-parallel curve-verified leaders until an official TE Curve Verification Pipeline refresh is run and accepted.

Acceptance Decision

Stage 1 is accepted as a polished-dataset training smoke pass because:

- every selected representative config completed in the accepted attempt;
- every accepted campaign leaderboard entry now records polished_dataset ;
- the schema recorded by accepted metrics is polished_point_v1 ;
- all accepted models use four input features and one TE target;
- the scalar winner is finite and materially improves the previous scalar registry level. The result is not a final full-program retraining conclusion. Stage 1 only validates the dataset integration and representative training paths.

Follow-Up

Recommended next steps

1. prepare the next polished-dataset retraining stage with larger and direction-separated queues;
2. keep the `TE Curve Verification Pipeline` refresh separate from normal campaign closeout;
3. after the next retraining stage, run curve-first verification on selected polished-trained candidates with explicit `global` , `Fw` , and `Bw` reporting.